

1 · CAUSALITY (Rubin potential outcomes)

$Y_i(1)$ = outcome if treated, $Y_i(0)$ = if not. $D_i=1$ treated. Observed: $Y_i = D_iY_i(1) + (1-D_i)Y_i(0)$.

Individual effect $\tau_i = Y_i(1) - Y_i(0)$ (heterogeneous: $\tau_i \neq \tau_j$)

ATE = $E[Y(1)-Y(0)]$ = avg τ over everyone = $E[Y(1)]-E[Y(0)]$

ATT = $E[Y(1)-Y(0) \mid D=1]$ = avg τ over treated only

Observed diff $E[Y(1)|D=1] - E[Y(0)|D=0]$

= **ATT + Selection bias** $E[Y(0)|D=1] - E[Y(0)|D=0]$

Selection bias = how treated would've done WITHOUT treatment vs. controls. $>0 \Rightarrow$ observed gap overstates effect.

Table recipe: (1) τ each row (2) ATE = mean τ all rows (3) Observed = mean $Y(1)$ of $D=1$ rows – mean $Y(0)$ of $D=0$ rows (4) ATT = mean τ of $D=1$ rows (5) Sel = mean $Y(0)$ of $D=1$ – mean $Y(0)$ of $D=0$ (6) CHECK ATT+Sel = Observed. Slides ex: 35.75 = 20.75 + 15; ATE=17.5

Fundamental problem of causal inference (Holland 1986): observe at most ONE potential outcome per unit at a time; the other is the **counterfactual** (missing). Clones = perfect control (thought experiment).

RCT: random assignment $\Rightarrow D \perp (Y(0), Y(1)) \Rightarrow E[Y(0)|D]=E[Y(0)|D=0] \Rightarrow$ **selection = 0** $\Rightarrow \hat{Y}_T - \hat{Y}_C$ estimates **ATE** (=ATT). Control group stands in for treated's missing $Y(0)$. Balances traits in *expectation* (not clones; small samples can be unbalanced by chance).

RCT limits: noncompliance, attrition, spillovers, subjects know they're studied, small-sample imbalance, cost/ethics/infeasible, **external validity**, short run only.

No RCT? Natural expt / quasi-expt / observational. Draft lottery (Angrist '90), quarter of birth IV (Angrist-Krueger '91), Indonesia schools (Duflo '01), Mariel boatlift (Card '90), NJ vs PA min wage DiD (Card–Krueger '94), close elections RD (Lee '08), Oregon Medicaid lottery. LaLonde '86: observational methods failed to match RCT.

Why X, Y correlate: $X \rightarrow Y$ causal; **reverse** $Y \rightarrow X$; **confounder** $Z \rightarrow$ both; selection; common time **trends**; chance/data mining (spurious: margarine–divorce $r=.99$). Correlation = symmetric, descriptive; causation = directional, needs counterfactual + design.

Prediction vs policy: X can predict Y well as a *proxy* for confounders; intervening on X doesn't move the confounders $\Rightarrow$ good forecast $\neq$ causal effect.

Deterministic $Y=f(X)$ exactly ($F=9/5 \cdot C+32$; $\text{tax}=\tau \cdot \text{price}$; $\text{dist}=\text{speed} \cdot \text{time}$; $C=30+8X$). **Stochastic** $Y=\beta_0+\beta_1X+u$ (wage on educ): same X , different Y . Econometrics = stats + econ theory + math (Frisch 1933).

Data structures

Type	What	+ / –
Cross section	many units, ONE time (Y_i)	+big, rich variation / –selection, OVB, timing unclear
Time series	ONE unit/aggregate, many times (Y_t) e.g. GDP	+dynamics, forecasting / –serial dependence, spurious trends, breaks
Repeated cross sec. (pooled)	NEW sample each period	+tracks population change / –can't see individual transitions; composition shifts
Panel (longitudinal)	SAME units, many times (Y_{it})	+ within-unit change , fixed effects remove time-invariant traits / –attrition, complex

“Same people revisited” = panel. “Fresh sample each year/month” = repeated cross section. Only panel shows **within-person/district change**.

2 · PROBABILITY & STATS

RV maps outcomes $\rightarrow$ numbers ($X: \Omega \rightarrow \mathbb{R}$). Capital X = random; lowercase x = realized. **Discrete** (countable) vs **continuous** (interval).

PMF $p(x)=P(X=x)$; valid iff all $p \geq 0$ AND $\sum p = 1$. **CDF** $F(x)=P(X \leq x)=\sum_{t \leq x} p(t)$. Discrete CDF = **step function**: flat between values, jumps = $p(x)$ at each value; 0 left of min, 1 at/after max; never decreases.

$P(X>x)=1-F(x)$; $P(a<X \leq b)=F(b)-F(a)$; $P(X \geq 2)=1-P(X \leq 1)$.

$E[X] = \sum x \cdot p(x)$ $E[X^2] = \sum x^2 \cdot p(x)$
 $\text{Var}(X) = E[(X-\mu)^2] = E[X^2] - (E[X])^2$ $SD = \sqrt{\text{Var}}$
 $E[a+bX] = a+bE[X]$; $\text{Var}(a+bX) = b^2\text{Var}(X)$;
 $SD(a+bX)=|b|SD$
 $\text{Var}(X \pm Y) = \text{Var}X + \text{Var}Y \pm 2\text{Cov}$; indep $\Rightarrow \text{Cov}=0$
Indicator $D \in \{0,1\}$: $E[D]=P(D=1)=p$, $\text{Var}=p(1-p)$

Units: E , SD , median = units of X ; Var = units². +a shifts center only (spread unchanged).

Mean (uses all, outlier-sensitive), **median** (middle; robust), **mode** (most frequent). Right skew $\Rightarrow$ mean $>$ median.

Sample statistics (use n–1)

$\bar{X} = \sum x_i/n$; $s^2 = \sum (x_i - \bar{x})^2/(n-1)$; $s = \sqrt{s^2}$. $n-1$ because $\bar{x}$ used up 1 df (corrects downward bias). Population: μ , σ^2 , ρ , N (fixed, unknown). Sample: $\bar{X}$, s^2 , r , n (random).

Rescale data by c (e.g. min $\rightarrow$ sec $c=60$): mean $\times c$, median $\times c$, $SD \times c$, **var** $\times c^2$. Add a: mean $+a$, var same.

Covariance & correlation

$\text{Cov}(X, Y) = E[(X-\mu_X)(Y-\mu_Y)] = E[XY] - E[X]E[Y]$; sample $s_{xy} = \sum (x-\bar{x})(y-\bar{y})/(n-1)$. $\text{Cov}(X, X) = \text{Var } X$. $\text{Cov}(a+bX, c+dY) = bd \cdot \text{Cov}$ (unit-dependent).

Corr $\rho = \text{Cov}/(\sigma_X \sigma_Y)$, $\in [-1,1]$, unit-free, symmetric; mult. one var by negative flips sign. **Corr=0 $\neq$ no relationship**: X symmetric about 0, $Y=X^2 \Rightarrow \text{Cov}=E[X^3]=0$ but perfectly dependent (only LINEAR assoc. measured). $\text{Corr} \neq$ causation.

Conditional probability / joint tables

$P(A|B) = P(A \cap B)/P(B)$ — denominator = **reference group** (“among B cases...”). Marginals = row/col totals. $P(A|B) \neq P(B|A)$ generally.

Independent iff $P(A \cap B) = P(A)P(B) \Rightarrow P(A|B) = P(A)$. Show one numeric comparison. Indicators R, L : $E[RL] = P(R=1, L=1)$; $\text{Cov}(R, L) = P(\text{both}) - P(R)P(L)$. $\text{Cov}=0$ = occur together, NOT causal.

Prediction w/ 0.5 cutoff: predict yes if $P>.5$; accuracy = share correct. Conditioning on informative var improves forecasts.

3 · CONTINUOUS & NORMAL

PDF $f(x) \geq 0$, total area 1; **$P(a \leq X \leq b)$ = area** = $\int_a^b P(X=c)=0$ (point has zero width) but interval has positive area = $f \cdot \text{width}$; density height can exceed 1 (not a prob). Endpoints don't matter: $P(a < X < b) = P(a \leq X \leq b)$.

$X \sim N(\mu, \sigma^2)$ — 2nd number is VARIANCE ($N(20,5^2) \Rightarrow \sigma=5$)
 $Z = (X-\mu)/\sigma \sim N(0,1)$; z = # SDs from mean (unit-free, compare across dists)
 $P(X \leq b) = \Phi(z_b)$; $P(X > b) = 1 - \Phi(z_b)$; **$P(a \leq X \leq b) = \Phi(z_b) - \Phi(z_a)$**
 $\Phi(-z) = 1 - \Phi(z)$; $P(-k \leq Z \leq k) = 2\Phi(k) - 1$; symmetric, mean=median=mode

68–95–99.7: $\mu \pm 1\sigma \approx 68\%$ (16% each tail), $\mu \pm 2\sigma \approx 95\%$ (2.5% each tail), $\mu \pm 3\sigma \approx 99.7\%$ (.15%). Central 95% = $[\mu-2\sigma, \mu+2\sigma]$ (exact $\pm 1.96\sigma$).

$\Phi(z)$	.0	.1	.2	.3	.4	.5	.6	.7	.8	.9
0.	.5000	.5398	.5793	.6179	.6554	.6915	.7257	.7580	.7881	.8159
1.	.8413	.8643	.8849	.9032	.9192	.9332	.9452	.9554	.9641	.9713
2.	.9772	.9821	.9861	.9893	.9918	.9938	.9953	.9965	.9974	.9981

$\Phi(3) = .9987$ $\Phi(1.645) = .95$ $\Phi(1.96) = .975$ $\Phi(2.576) = .995$. Ex: $P(70 \leq X \leq 85)$, $X \sim N(75, 10^2)$: $\Phi(1) - \Phi(-.5) = .8413 - .3085 = .5328$

4 · ESTIMATORS

Estimator $\hat{\theta} = g(X_1, \dots, X_n)$ = a RULE, random variable w/ sampling distribution. **Estimate** = the NUMBER from one observed sample. Parameter θ fixed, unknown.

Bias = $E[\hat{\theta}] - \theta$ (unbiased: $E[\hat{\theta}] = \theta$, cluster centered on bullseye)
Efficiency: among UNBIASED estimators, smaller Var wins
Consistent: $P(|\hat{\theta}_n - \theta| > \epsilon) \rightarrow 0$ as $n \rightarrow \infty$. Check: **Bias $\rightarrow 0$ AND Var $\rightarrow 0$**
MSE = $E[(\hat{\theta} - \theta)^2] = \text{Var}(\hat{\theta}) + \text{Bias}^2$ (small bias OK if Var drops a lot; ranking depends on true θ)

Tools (iid): $E[\bar{X}] = \mu$, **$\text{Var}(\bar{X}) = \sigma^2/n$** , $SD(\bar{X}) = \sigma/\sqrt{n}$; $E[\Sigma X] = n\mu$, $\text{Var}(\Sigma X) = n\sigma^2$. $E[c\hat{\theta}] = cE[\hat{\theta}]$, $\text{Var}(c\hat{\theta}) = c^2\text{Var}(\hat{\theta})$. Constants add to E only.

Bernoulli(p): $E[X_i] = p$, $\text{Var}=p(1-p)$. $S = \Sigma X_i$; **$E[S] = np$, $\text{Var}(S) = np(1-p)$** .
General: $\hat{\theta} = (aS+b)/c \Rightarrow E = (a \cdot np + b)/c$, $\text{Var} = a^2 \cdot np(1-p)/c^2$.

Estimator	E	Bias	Var	Cons?
$\hat{p} = S/n$	p	0	$p(1-p)/n$	Yes
$\hat{p} = (S+1)/(n+2)$	$(np+1)/(n+2)$	$(1-2p)/(n+2)$	$np(1-p)/(n+2)^2$	Yes
$\hat{p}_1 = X_1$	p	0	$p(1-p)$	No
$\bar{X}$	μ	0	σ^2/n	Yes
X_1	μ	0	σ^2 (ignores extra data)	No
$k\bar{X}$, $k=n/(n+1)$	$k\mu$	$-\mu/(n+1)$	$k^2\sigma^2/n = \sigma^2/(n+1)^2$	Yes

$MSE(n/(n+1) \cdot \bar{X}) = (n\sigma^2 + \mu^2)/(n+1)^2$. Add-two shrinks toward .5 $\Rightarrow$ lower Var, biased unless $p=.5$; $p=.52$, $n=9$: $MSE .0186 < .0277$ (S/n). $p=.52$, $n=11$: $.0163 < .0227$. $\bar{X}$ more efficient than X_1 for $n>1$.

Median/mode of 0/1 data = $1\{S>n/2\}$ (odd n): estimates the **majority outcome** $1\{p>.5\}$, NOT the size of p ; $\rightarrow 1$ if $p>.5 \Rightarrow$ inconsistent for p .

5 · SIMPLE REGRESSION (OLS)

Population: $Y = \beta_0 + \beta_1X + u$. Y dependent/outcome/LHS; X regressor/RHS. β 's fixed, unknown. u = all other determinants + nonlinearity + measurement error (unobserved).

Zero conditional mean $E[u|X=x]=0$: among units with same X , unobservables average 0 $\Rightarrow E[Y|X=x] = \beta_0 + \beta_1x$ (line = conditional mean); β_1 = change in $E[Y|X]$ per +1 X . $u \perp X$ & $E[u]=0$ is stronger/sufficient. **NOT needed to CALCULATE OLS** — only to INTERPRET (only need variation in X).

OVB story: income $\in u$, affects score, correlated w/ class size $\Rightarrow E[u|\text{STR}] \neq 0 \Rightarrow$ slope mixes STR effect with income $\Rightarrow$ not causal (still valid description).

OLS: choose b_0, b_1 to min $Q = \text{SSR} = \sum (y_i - b_0 - b_1x_i)^2$. Why squares: $\Sigma 0$ cancels (+8, –8); squares smooth/differentiable, closed form, penalize big misses, target cond. mean. $|\hat{u}| \rightarrow$ median regression, not diff. at 0.

FOCs: $\partial Q/\partial b_0 = -2\sum (y_i - b_0 - b_1x_i) = 0 \Rightarrow \Sigma \hat{u}_i = 0 \Rightarrow \bar{y} = \hat{\beta}_0 + \hat{\beta}_1\bar{x} \Rightarrow$ line passes through **$(\bar{x}, \bar{y})$** .

$\partial Q/\partial b_1 = -2\sum x_i(y_i - b_0 - b_1x_i) = 0 \Rightarrow \Sigma x_i\hat{u}_i = 0$. These are algebraic sample identities; they hold always and do NOT prove population $E[u|X]=0$.

$$\hat{\beta}_1 = \frac{\Sigma(x_i - \bar{x})(y_i - \bar{y})}{\Sigma(x_i - \bar{x})^2} = \frac{s_{xy}/s_x^2}{s_x^2/s_x^2} = \frac{r_{xy} \cdot s_y/s_x}{s_x^2/s_x^2} = \frac{(S_{xy} - n\bar{x}\bar{y})/(Sx^2 - n\bar{x}^2)}{\hat{\sigma}_x^2} \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x}$$

$\text{sign}(\hat{\beta}_1) = \text{sign}(r) = \text{sign}(\text{Cov})$ since $s_x, s_y > 0$. **Residual** $\hat{u}_i = y_i - \hat{y}_i$ (units of Y ; + = point above line).

Hand-calc table: $x \mid y \mid x-\bar{x} \mid y-\bar{y} \mid (x-\bar{x})^2 \mid (x-\bar{x})(y-\bar{y}) \rightarrow$ sum last two $\rightarrow \hat{\beta}_1$, $\hat{\beta}_0 \rightarrow \bar{y} = \hat{\beta}_0 + \hat{\beta}_1\bar{x} \rightarrow \hat{u} = y - \hat{y} \rightarrow$ check $\Sigma \hat{u} = 0$, $\Sigma x\hat{u} = 0$ (rounding ≈ 0). Ex (1,2)(2,3)(3,5): $\hat{\beta}_1 = 1.5$, $\hat{\beta}_0 = 1/3$.

Change	New $\hat{\beta}_1$	New $\hat{\beta}_0$
$y^* = c \cdot y$	$c \cdot \hat{\beta}_1$	$c \cdot \hat{\beta}_0$
$x^* = c \cdot x$	$\hat{\beta}_1/c$	same
$y^* = y + a$	same	$\hat{\beta}_0 + a$
$x^* = x + a$	same	$\hat{\beta}_0 - \hat{\beta}_1 a$

Why: $\Sigma(x-\bar{x})(y-\bar{y})$ scales by $c_x c_y$, $\Sigma(x-\bar{x})^2$ by c_x^2 . Units only relabel — same fit. $\$$ = cents: slope $\times 100$.

Interpretation templates

Slope: “A one-[unit X] increase in X is **associated with** a $\hat{\beta}_1$ [units of Y] higher/lower Y , on average.” (units = Y per unit X)

Intercept: fitted Y when $X=0$; if 0 outside data range $\Rightarrow$ extrapolation, little meaning.

Prediction: $\hat{y}(x_i) = \hat{\beta}_0 + \hat{\beta}_1 x_{i0}$ estimates $E[Y|X=x_{i0}]$ (avg for such units, not each one). Change: $\Delta \hat{y} = \hat{\beta}_1 \cdot \Delta x$. Beware x_{i0} outside observed range.

Not causal because: no random assignment; selection/confounders in u (ability, motivation, income); reverse causality; small n .

Reading R output

Intercept & slope “Estimate” column $\Rightarrow \hat{y} = b_0 + b_1x$. **Residual df** = $n - 2$ (2 coeffs) $\Rightarrow n = \text{df} + 2$. Class-size: Score[^] = 698.93 – 2.28 · STR, df 418 $\Rightarrow n=420$; STR 22 $\rightarrow$ 18: +9.12 pts (not proven causal).

6 · EXAM TACTICS

Write formula $\rightarrow$ plug in $\rightarrow$ answer with **units** $\rightarrow$ one-sentence interpretation. Check PMF sums to 1 & $\Sigma \hat{u} = 0$. Name the **reference group** for conditional probs. For any “does this prove causation?” $\rightarrow$ **No** + name a confounder/selection/reverse causality + what design fixes it (RCT). Distinguish estimator vs estimate, population vs sample, u vs $\hat{u}$.

7 · SYMBOL GLOSSARY

Causality: $Y_i(1)$ outcome for unit i if treated (potential outcome) · $Y_i(0)$ outcome for unit i if NOT treated · Y_i outcome actually observed for i · D_i treatment indicator: 1 = treated, 0 = control · τ_i (**tau**) individual treatment effect $Y_i(1)-Y_i(0)$ · **ATE** avg treatment effect, whole population · **ATT** avg treatment effect on the treated ($D=1$) · **RCT** randomized controlled trial · $\hat{Y}_T, \hat{Y}_C$ sample mean outcome, treatment / control group · n_T, n_C # in treatment / control group · **I** or **I.L** “is independent of” · **IV, DiD, RD** instrumental variables, difference-in-differences, regression discontinuity (quasi-expt designs) · i , t index for unit (person/district) and time period

Probability & statistics: **X (capital)** random variable (value not yet known) · **x (lowercase)** a realized/observed value of X · **Ω , ω** set of all possible outcomes; one outcome · **$p_X(x)$** PMF = $P(X=x)$ (discrete) · **$f_X(x)$** PDF = density height (continuous; not a prob) · **$F_X(x)$** CDF = $P(X \leq x)$ · **$P(A \cap B)$** prob. A AND B both happen · **$P(A|B)$** prob. of A GIVEN B happened · **$E[X]$, μ (**mu**)** expected value / population mean · **$\text{Var}(X)$, σ^2** population variance · **σ (**sigma**)**, **SD** population standard deviation · **$\text{Cov}(X, Y)$, σ_{xy}** population covariance · **Corr, ρ (**rho**)** population correlation · **N , n** population size, sample size · **$\bar{X}$, $\bar{x}$** sample mean (“ X -bar”) · **s^2 , s** sample variance, sample SD ($n-1$) · **s_{xy} , r_{xy}** sample covariance, sample correlation · $\sim$ “is distributed as” · **$N(\mu, \sigma^2)$** normal dist. with mean μ , VARIANCE σ^2 · **Z , z** standard normal $N(0,1)$ variable; z -score · **$\Phi(z)$ (**phi**)** standard normal CDF = $P(Z \leq z)$ · **Bernoulli(p)** 0/1 variable, $P(=1) = p$ · **$1\{\cdot\}$** indicator: 1 if condition true, else 0 · **iid** independent & identically distributed · **$\Sigma, \int$** sum; integral (area under curve)

Estimators: **θ (**theta**)** generic unknown population parameter · **$\hat{\theta}$** estimator of θ (“theta-hat”) · **$g(X_1, \dots, X_n)$** the rule/function that makes the estimator · **p** true population probability/share · **$S \Sigma X_i = \#$ of 1s (successes) in sample** · **$\hat{p}$** S/n , sample proportion · **$\hat{p} = (S+1)/(n+2)$** , “add-two” (p -tilde) · **p_1, X_1** just the first observation · **X_i** observation i in the sample · **ϵ (**epsilon**)** any small positive number $\rightarrow p$ converges in probability (consistency) · **Bias, MSE** $E[\hat{\theta}] - \theta$; mean squared error

Regression: **Y** dependent var. / outcome / LHS · **X** independent var. / regressor / RHS · **β_0 (**beta**)** population intercept (unknown) · **β_1** population slope (unknown) · **u** population error: all other factors (unobserved) · **$\hat{\beta}_0, \hat{\beta}_1$** OLS estimates of intercept, slope · **$\hat{Y}_i, \hat{y}_i$** fitted/predicted value on the line · **$\hat{u}_i$** residual = $y_i - \hat{y}_i$ (sample, observed) · **$E[Y|X=x]$** conditional mean of Y among units with $X=x$ · **$E[u|X]=0$** zero conditional mean assumption · **b_0, b_1** any candidate intercept/slope · **Q , SSR** sum of squared residuals $\sum (y_i - b_0 - b_1x_i)^2$ · **arg min** the values that minimize · **$\partial Q/\partial b$** partial derivative (FOC: set = 0) · **OLS** ordinary least squares · **df** residual degrees of freedom = $n - 2$ · **STR** student–teacher ratio (class size) · **OVB** omitted variable bias · **$\text{lm}(Y \sim X)$** R command: regress Y on X